

Incident Response Report

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Environment: Wazuh + Metasploit + MITRE Caldera SOC Lab

1. Executive Summary

On 14th April, 2026, a simulated cyberattack was executed in a controlled SOC lab using Metasploit and MITRE Caldera to emulate real-world adversary behavior. Wazuh successfully detected suspicious activities including port scanning, SMB exploitation attempts, and lateral movement techniques. The attack targeted a vulnerable Metasploitable2 machine exploiting the Samba usermap script vulnerability. Alerts generated by Wazuh enabled rapid detection and response. The malicious activity was contained by blocking the attacker IP using iptables. No real data loss occurred as this was a controlled environment. This exercise validated the effectiveness of detection, triage, and response capabilities.

2. Timeline of Events

Time	Event Description
10:05 AM	Wazuh alert triggered for suspicious SMB traffic
10:07 AM	Metasploit port scan and service enumeration detected
10:08 AM	Samba usermap_script exploit launched
10:10 AM	Unauthorized shell access attempt observed
10:12 AM	Attacker IP blocked via iptables
10:14 AM	MITRE Caldera executed additional adversary techniques
10:20 AM	System monitored and restored to normal state

3. Root Cause Analysis (RCA)

The root cause of the incident was the presence of a known vulnerability in the Samba service running on the Metasploitable2 system. The system was intentionally left unpatched for testing purposes. Lack of segmentation and exposed services allowed exploitation. Additionally, weak monitoring configurations prior to tuning could have delayed detection in a real-world scenario.

4. Recommendations

- Regularly update and patch all services to mitigate known vulnerabilities.
- Implement network segmentation to isolate critical systems.
- Enhance Wazuh rules for better detection of SMB and exploitation attempts.
- Use MITRE Caldera for continuous adversary emulation and testing.
- Deploy firewall rules and automated blocking mechanisms (e.g., CrowdSec).
- Continuously monitor logs and integrate with SIEM for real-time alerts.